The second minimum of Barnes-Wall lattices

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In memory of Jacques Martinet

Abstract. The second minimum of the Barnes-Wall lattices is a least 3/2 of the minimum.
Les reseaux de Barnes-Wall ne possèdent aucun vecteur de norme a avec $d < a < 3d/2$, où d est le minimum du reseau.
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1 Introduction

The Barnes-Wall lattices form an infinite series of lattices of dimension 2^m for $m \in \mathbb{N}$. They are constructed in [2] which explicitly elaborates a subset of their minimal vectors to show that the Barnes-Wall lattices are locally densest lattices. This fact also follows from an inspection of their automorphism group. For $m \neq 3$ this group is a subgroup of index 2 of the real Clifford group (see [10]), a fact that allows Bachoc in [1] to show that for $m \geq 3$ all non-empty layers of the Barnes-Wall lattices form spherical 6-designs. This paper gives a construction of the Barnes-Wall lattices as subdirect products (Theorem 3.2) that is used to give an easy proof of their kissing number (Lemma 3.5) and to show that for $m \geq 3$ the minimal vectors form a spherical 6-design (Proposition 3.7). This construction also allows to derive the main result of this paper, Theorem 3.3, showing that the Barnes-Wall lattices of minimum d have no vectors of norm a with $d < a < 3d/2$.

The most prominent construction of the Barnes-Wall lattices is by applying Construction D to a chain of Reed-Muller codes (see for instance [6, Chapter 8, Section 8], [2], and also [8] for a basis-independent formulation). Berlekamp and Sloane [3] show that in the r-th order binary Reed-Muller code of length 2^m and

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minimum distance $d = 2^{m-r}$, the only codewords having weight between d and $2d$ are those with weights of the form $2d - 2^i$ for some i. Motivated by this observation, certain experiments, and the theta series of the Barnes-Wall lattice of dimension 64 and 128 in [12], Christoph Keller and his PhD student Ashley Roberts [9] conjectured that an analogous property should also be true for the Barnes-Wall lattices. In the meantime this conjecture has been proven in [5].

In [8] we show that Barnes-Wall lattices are obtained by applying Construction A to Reed-Muller codes over chain rings. Knowing the weight distribution of these chain ring codes could give an alternative approach to obtain the theta series of the Barnes-Wall lattices.

2 The two Barnes-Wall lattices of dimension 2^m

In [2] Barnes and Wall construct a series of lattices in dimension $N := 2^m$ for any $m \in \mathbb{N}$. Put $\mathcal{V}_m = \mathbb{F}_2^m$ to denote the m-dimensional vector space over the field with 2 elements and fix a basis $(v_1, \ldots, v_m)$ of $\mathcal{V}_m$. Put

$$\mathcal{T}_r(m) := \{\mathcal{U} \subseteq \mathcal{V}_m \mid \mathcal{U} = \langle v_i \mid i \in I\rangle_{\mathbb{F}_2}, I \subseteq \{1, \ldots, m\}, |I| = r\} \tag{2.1}$$

to denote the set of r-dimensional subspaces of $\mathcal{V}_m$ that have a basis that is a subset of $\{v_1, \ldots, v_m\}$.

We now let $\mathcal{V}_m$ index the elements of an orthonormal basis

$$(e_v \mid v \in \mathcal{V}_m) \tag{2.2}$$

of the euclidean space $(\mathbb{R}^N, (\ ,\ ))$. Let

$$\Gamma_m := \langle e_v \mid v \in \mathcal{V}_m\rangle_{\mathbb{Z}}$$

denote the standard lattice spanned by this orthonormal basis. For any subset $\mathcal{U}$ of $\mathcal{V}_m$ we put

$$x_{\mathcal{U}} := \sum_{v \in \mathcal{U}} e_v \in \Gamma_m.$$

Then [2] defines the following sublattices of Γ_m:

Definition 2.1. *Let $\lambda := (\lambda_0, \ldots, \lambda_m) \in \mathbb{Z}^{m+1}$ such that $\lambda_0 = 0, \lambda_r - 1 \leq \lambda_{r-1} \leq \lambda_r$ for all $1 \leq r \leq m$. Then*

$$\Lambda(\lambda) := \langle 2^{\lambda_{m-r}} x_{\mathcal{U}} \mid 0 \leq r \leq m, \mathcal{U} \text{ affine subspace of } \mathcal{V}_m, \dim(\mathcal{U}) = r\rangle_{\mathbb{Z}}.$$

For λ as in Definition 2.1 put $\lambda' := (\lambda'_0, \ldots, \lambda'_m)$ where $\lambda'_r = \lambda_m - \lambda_{m-r}$. Then [2, Theorem 3.1, Theorem 3.2] give the following properties of the lattices $\Lambda(\lambda)$ and their minimum

$$\min(\Lambda) = \min\{(x, x) \mid 0 \neq x \in \Lambda\}.$$

Proposition 2.2. *(a) A $\mathbb{Z}$-basis of $\Lambda(\lambda)$ is given by*

$$\bigcup_{r=0}^{m}\{2^{\lambda_{m-r}}x_{\mathcal{U}} \mid \mathcal{U}\in\mathcal{T}_r(m)\}$$

where $\mathcal{T}_r(m)$ is given in (2.1).

(b) $\det(\Lambda(\lambda))=2^{2d}$ where $d=\sum_{r=0}^{m}\lambda_r\binom{m}{r}$.

(c) $\min(\Lambda(\lambda))=2^{\alpha}$ where $\alpha=\min\{m-r+2\lambda_r \mid 0\le r\le m\}$.

Barnes and Wall single out two particular lattices among the lattices $\Lambda(\lambda)$:

Theorem 2.3. *([2, Theorem 4.3]) Put $\lambda_r:=\lfloor\frac{r}{2}\rfloor$ and $\mu_r:=\lfloor\frac{r+1}{2}\rfloor$ $(0\le r\le m)$ and put*

$$\Lambda_m:=\Lambda(\lambda),\ \Delta_m:=\Lambda(\mu).$$

The index of Λ_m in the standard lattice Γ_m is 2^j where

$$j=\log_2([\Gamma_m:\Lambda_m])=\sum_{r=0}^{m}\lfloor\frac{r}{2}\rfloor\binom{m}{r}=(m-1)2^{m-2}.$$

Then $\Lambda_m\supset\Delta_m\supset 2\Lambda_m$ with $\log_2([\Lambda_m:\Delta_m])=\log_2([\Delta_m:2\Lambda_m])=2^{m-1}$.
For the minimum of the two lattices we get $\min(\Lambda_m)=2^{m-1},\min(\Delta_m)=2^m$.

Remark 2.4. (see [2, Theorem 4.3] and its proof)
The two lattices Λ_m and Δ_m are similar.
More precisely the linear map σ defined on the orthonormal basis from (2.2) by

$$\sigma(e_v):=\begin{cases} e_v+e_{v+v_m} & \text{if } v\in\langle v_1,\dots,v_{m-1}\rangle \\ -e_v+e_{v+v_m} & \text{if } v\notin\langle v_1,\dots,v_{m-1}\rangle\end{cases}$$

satisfies $(\sigma(x),\sigma(y))=2(x,y)$ for all $x,y\in\mathbb{R}^N$ and $\sigma(\Lambda_m)=\Delta_m$.

The lattices that are commonly known as "the" Barnes-Wall lattices are scaled versions of the lattices Λ_m from Theorem 2.3:

Remark 2.5. If $m\ge 3$ is odd then $\mathrm{BW}_m:=2^{-(m-1)/4}\Lambda_m$ is an even unimodular lattice of minimum $\min(\mathrm{BW}_m)=2^{(m-1)/2}$.
If m is even then $\mathrm{BW}_m:=2^{-(m-2)/4}\Lambda_m$ is an even 2-modular lattice of minimum $\min(\mathrm{BW}_m)=2^{m/2}$.

For $m\ne 3$ we denote by $G_m:=\mathrm{Aut}(\Lambda_m)$ the automorphism group of the lattice Λ_m. For $m=3$ the lattice Λ_m is similar to the root lattice E_8 and has a larger automorphism group. To get a unified notation we put G_3 to denote the stabiliser of Δ_3 in $\mathrm{Aut}(\Lambda_3)$. Then

$$G_m\cong 2_+^{1+2m}.\Omega_{2m}(2)$$

is an extension of the extraspecial group of order 2^{1+2m} that is the central product (or tensor product) of m copies of the dihedral group of order 8 with the orthogonal group $\Omega_{2m}(2)$ of a split quadratic space of dimension $2m$ over $\mathbb{F}_2$. The group G_m is a normal subgroup of index 2 in the real Clifford group $\langle G_m, h\rangle \leq \mathrm{GL}_N(\mathbb{R})$ (see for instance [1], [10]). The element $\sigma = \sqrt{2}h \in \mathrm{GL}_N(\mathbb{Q})$ induces a similarity between Λ_m and $\Delta_m = \sigma(\Lambda_m)$. The group G_m is studied in detail in [4].

Theorem 2.6. *([4, Théorème II.4])*

(a) Both lattices Λ_m and Δ_m are invariant under G_m.

(b) G_m acts transitively on the set of minimal vectors of Λ_m and of Δ_m.

3 The second minimum of the lattices Λ_m

3.1 The minimal classes in Λ_m/Δ_m

The key observation for having a recursive proof of the second minimum of the Barnes-Wall lattices is given in the following lemma.

Lemma 3.1. *Let $\ell \in \Lambda_m$ be a minimal vector, i.e. $(\ell, \ell) = 2^{m-1}$. Then for any $x \in \ell + \Delta_m$ either $(x, x) = (\ell, \ell) = 2^{m-1}$ or $(x, x) \geq 2^m$.*

Proof. By Theorem 2.6 the group G_m acts transitively on the set of minimal vectors of Λ_m and hence also on the classes of Λ_m/Δ_m that contain such a minimal vector. So we make a suitable choice of the minimal vector ℓ.
First assume that m is odd. Then we choose the minimal vector

$$\ell = 2^{(m-1)/2}e_0 \in \Lambda_m$$

and let

$$x = \ell + d \in \ell + \Delta_m \text{ with } (x, x) > 2^{m-1}.$$

Write $x = \sum_{v\in\mathcal{V}_m} a_v e_v$ with coefficients $a_v \in \mathbb{Z}$ in the orthonormal basis from (2.2) and let 2^j be the maximal 2-power that divides all a_v.
If $j \geq (m-1)/2$ then (x, x) is a multiple of 2^{m-1} and hence $(x, x) \geq 2^m$ by the assumption that $(x, x) > 2^{m-1}$.
So assume that $j < (m-1)/2$ and put

$$y := 2^{-j}x = \sum_{v\in\mathcal{V}_m} b_v e_v$$

with $b_v = 2^{-j}a_v \in \mathbb{Z}$. Then

$$2^{-j}d = (b_0 - 2^{(m-1)/2-j})e_0 + \sum_{0\neq v\in\mathcal{V}_m} b_v e_v \in 2^{-j}\Delta_m \cap \Gamma_m$$

and the set $S := \{v \in \mathcal{V}_m \mid b_v \text{ odd }\}$ is the set of indices of the odd coefficients in $2^{-j}d$. As in the proof of [2, Theorem 3.2] we see that $2^{-j}d$ lies in the lattice called Λ_{2j} in [2, Lemma 3.3]. So [2, Lemma 3.3] implies that the cardinality of S is $\geq 2^{m-2j}$ and hence

$$(x,x) = 2^{2j}(y,y) \geq 2^{2j}|S| \geq 2^m.$$

This shows the lemma if m is odd.
For even m, there are no minimal vectors of Λ_m that are scalar multiples of one of the e_v. However, as Λ_m and Δ_m are similar (see Remark 2.4), we may use the same argument as before for the minimal classes of $\Delta_m/2\Lambda_m$.
So we choose the minimal vector

$$\ell = 2^{m/2}e_0 \in \Delta_m$$

and assume that $d \in 2\Lambda_m$ is such that

$$x = \ell + d \in \ell + 2\Lambda_m \text{ has norm } (x,x) > 2^m = \min(\Delta_m).$$

Then the same argument as before shows that $(x,x) \geq 2^{m+1}$ which shows the lemma also for m even. □

3.2 A recursive construction of Λ_m as a subdirect product

The Barnes-Wall lattices have an easy construction as a subdirect product, very similar to the doubling construction for the Reed-Muller codes. Fixing the basis $(v_1, \ldots, v_{m+1})$ of $\mathcal{V}_{m+1}$ as in (2.1) there is a natural embedding

$$\iota : \mathcal{V}_m = \langle v_1, \ldots, v_m \rangle \hookrightarrow \mathcal{V}_{m+1}, \iota(v) = v.$$

Combining ι with the translation along v_{m+1} we obtain a bijection

$$\tau : \mathcal{V}_m \to v_{m+1} + \iota(\mathcal{V}_m), v \mapsto v_{m+1} + \iota(v)$$

so that $\mathcal{V}_{m+1} = \iota(\mathcal{V}_m) \dot\cup \tau(\mathcal{V}_m)$. By abuse of notation we also use ι and τ to denote the $\mathbb{Z}$-linear maps

$$\iota : \Gamma_m \to \Gamma_{m+1}, e_v \mapsto e_{\iota(v)} \text{ for all } v \in \mathcal{V}_m$$

and

$$\tau : \Gamma_m \to \Gamma_{m+1}, e_v \mapsto e_{\tau(v)} \text{ for all } v \in \mathcal{V}_m.$$

Then ι and τ are isometric embeddings and Γ_{m+1} is the orthogonal sum of $\iota(\Gamma_m)$ and $\tau(\Gamma_m)$. In this notation we obtain

Theorem 3.2. $\Lambda_{m+1} = \{\iota(\ell) + \iota(d) + \tau(\ell) \mid \ell \in \Lambda_m, d \in \Delta_m\}$.

Proof. The right hand side, Λ, is a lattice, in fact we have

$$\Lambda = (\iota + \tau)(\Lambda_m) \oplus \iota(\Delta_m).$$

So the index of Λ in Γ_{m+1} is 2^j with

$$j = \log_2([\Gamma_{m+1} : \Lambda]) = \log_2([\Gamma_m : \Delta_m]) + \log_2([\Gamma_m : \Lambda_m]) = \\ (m+1)2^{m-2} + (m-1)2^{m-2} = m2^{m-1} = \log_2([\Gamma_{m+1} : \Lambda_{m+1}]).$$

So it remains to show that the basis of Λ_{m+1} given in Proposition 2.2 (a) is contained in Λ. So let $\mathcal{U} \in \mathcal{T}_r(m+1)$.
If $v_{m+1} \in \mathcal{U}$ then $\mathcal{U}' := \mathcal{U} \cap \mathcal{V}_m \in \mathcal{T}_{r-1}(m)$ and

$$2^{\lfloor \frac{m+1-r}{2} \rfloor} x_{\mathcal{U}} = \iota(2^{\lfloor \frac{m-(r-1)}{2} \rfloor} x_{\mathcal{U}'}) + \tau(2^{\lfloor \frac{m-(r-1)}{2} \rfloor} x_{\mathcal{U}'}) \in (\iota + \tau)(\Lambda_m).$$

If $v_{m+1} \notin \mathcal{U}$ then $\mathcal{U} \in \mathcal{T}_r(m)$ and

$$2^{\lfloor \frac{m+1-r}{2} \rfloor} x_{\mathcal{U}} = \iota(2^{\lfloor \frac{m-r+1)}{2} \rfloor} x_{\mathcal{U}'}) \in \iota(\Delta_m).$$

□

For related constructions see for instance [7] and [11].

3.3 The main result

Theorem 3.3. *Let $x \in \Lambda_m$ be such that $(x,x) > \min(\Lambda_m) = 2^{m-1}$. Then $(x,x) \geq 2^{m-1} + 2^{m-2}$.*

Proof. We proceed by induction on m. The cases $m = 2, \ldots, 5$ follow immediately from Remark 2.5, as here BW_m is an even lattice of minimum ≤ 4.

For the induction step assume that the theorem holds for m and let $x = \iota(\ell) + \iota(d) + \tau(\ell) \in \Lambda_{m+1}$ with

$$2^m < (x,x) = (\ell + d, \ell + d) + (\ell, \ell) < 2^m + 2^{m-1}.$$

Without loss of generality we assume that $(\ell, \ell) \leq (\ell + d, \ell + d)$.
Then there are three cases to consider:
If $\ell = 0$, then $d \in \Delta_m$ is not a minimal vector as $(x,x) = (d,d) > 2^m$. Hence by induction hypothesis (using Remark 2.4) we have $(x,x) = (d,d) \geq 2^m + 2^{m-1}$.
For $0 \neq \ell \in \Lambda_m$ not a minimal vector by assumption $(\ell, \ell) \geq 2^{m-1} + 2^{m-2}$ and hence $(x,x) \geq 2(\ell, \ell) \geq 2^m + 2^{m-1}$.
In the last case $\ell \in \Lambda_m$ is a minimal vector, i.e. $(\ell, \ell) = 2^{m-1}$. By assumption $(x,x) > 2^m$, therefore $\ell + d \in \ell + \Delta_m$ is not a minimal vector, so $(\ell + d, \ell + d) \geq 2^m$ by Lemma 3.1. Therefore $(x,x) \geq 2^m + 2^{m-1}$.
Combining these three cases shows the claim for $m+1$ and finishes the induction step. □

Remark 3.4. The construction of Λ_{m+1} from Theorem 3.2 yields an isometric embedding ι of Δ_m into Λ_{m+1}. As the second minimum of Δ_m is twice the second minimum of Λ_m, this gives an inductive proof that, for $m \geq 4$, the lattice Λ_{m+1} contains vectors of square length $2^m + 2^{m-1}$. So the bound in Theorem 3.3 is sharp if $m \geq 4$.

3.4 Some further consequences

This final section gives some simple proofs of well-known results based on Theorem 3.2. We first deduce an easy recursive formula for the set

$$S(\Lambda_m) := \{\ell \in \Lambda_m \mid (\ell, \ell) = 2^{m-1}\}$$

of minimal vectors in Λ_m and its cardinality $s_m := |S(\Lambda_m)| = |S(\Delta_m)|$ (see for instance [2, Section 5], [4, Corollaire 1]).

Lemma 3.5.

$$S(\Lambda_{m+1}) = \begin{array}{l} \{(\iota+\tau)(\ell) + \iota(d) \mid \ell, \ell + d \in S(\Lambda_m), d \in \Delta_m\} \cup \\ \{\iota(d) \mid d \in S(\Delta_m)\} \cup \{\tau(d) \mid d \in S(\Delta_m)\} \end{array}.$$

The kissing number s_{m+1} of Λ_{m+1} satisfies $s_1 = 4$ and for $m \geq 1$

$$s_{m+1} = (2^{m+1} + 2)s_m.$$

Proof. For any two minimal vectors $\ell_1, \ell_2 \in \Lambda_m$ with $(\ell_i, \ell_i) = 2^{m-1}$ and $\ell_1 + \Delta_m = \ell_2 + \Delta_m$ we have that $\ell_1 \pm \ell_2 \in \Delta_m$ and hence $2^m \pm 2(\ell_1, \ell_2) \geq 2^m = \min(\Delta_m)$. So $(\ell_1, \ell_2) = 0$ and the class $\ell_1 + \Delta_m$ contains at most 2^{m+1} minimal vectors, $\pm\ell_1, \pm\ell_2, \ldots, \pm\ell_{2^m}$ with $(\ell_i, \ell_j) = 0$ for $i \neq j$. To show that there are at least 2^{m+1} such minimal vectors in each class in Λ_m/Δ_m and hence also in $\Delta_m/2\Lambda_m$ we use Theorem 2.6 and Remark 2.4 to choose a suitable class. If m is odd put $\ell + \Delta_m = 2^{(m-1)/2}e_0 + \Delta_m$ and for even m choose the class $\ell + 2\Lambda_m = 2^{m/2}e_0 + 2\Lambda_m \in \Delta_m/2\Lambda_m$. These classes contain the 2^{m+1} minimal vectors $\pm 2^{\lfloor m/2 \rfloor}e_v$ for all $v \in \mathbb{F}_2^m$.
We now use the description of Λ_{m+1} from Theorem 3.2 to enumerate $S(\Lambda_{m+1})$:
We have seen that there are exactly $2^{m+1}s_m$ minimal vectors of the form $(\iota+\tau)(\ell) + \iota(d) \in \Lambda_{m+1}$, where $\ell, \ell + d \in \Lambda_m$ are both minimal vectors in the class $\ell + \Delta_m$. There are also $2s_m$ minimal vectors of the form $\iota(d)$ or $\tau(d)$ where $d \in \Delta_m$ is a minimal vector. Hence we have $s_{m+1} = (2^{m+1} + 2)s_m$. □

Using this description of the minimal vectors of Λ_m it is easy to show that for $m \geq 3$ these minimal vectors form a spherical 6-design. Though this is a well known fact (see for instance [1, Corollary 5.3]) the direct proof given below is substantially easier than the one using the description in [4, Théorème II.2] of the minimal vectors of the Barnes-Wall lattices.

Remark 3.6. (see [13, Théorèmes 3.2 and 8.1]) Let X be a finite subset of the euclidean space $(\mathbb{R}^N, (,))$ such that $(x,x) = M$ for all $x \in X$ and $-x \in X$ for all $x \in X$. For any natural number ℓ the set X is a spherical (2ℓ)-design if and only if one of the following two equivalent conditions is satisfied:

(a) For all $v \in \mathbb{R}^N$

$$\sum_{x\in X}(x,v)^{2\ell} = \frac{1\cdot 3\cdot 5\cdot \ldots \cdot (2\ell-1)}{N(N+2)\ldots(N+2(\ell-1))}(v,v)^\ell M^\ell |X|.$$

(b)

$$\sum_{x,y\in X}(x,y)^{2\ell} = \frac{1\cdot 3\cdot 5\cdot \ldots \cdot (2\ell-1)}{N(N+2)\ldots(N+2(\ell-1))}M^{2\ell}|X|^2$$

Using induction over m the construction in Theorem 3.2 allows us to give a quick proof of the following well known result.

Proposition 3.7. *For $m \geq 3$ the sets $S(\Lambda_m)$ and $S(\Delta_m)$ form spherical 6-designs.*

Proof. For $z \in \mathbb{R}$ and $m \in \mathbb{N}$ we put

$$c_m(z) := s_m \min(\Lambda_m)^3 z^3 \frac{1\cdot 3\cdot 5}{2^m(2^m+2)(2^m+4)}.$$

We proceed by induction on m, the case $m = 3$ can be settled by direct computations. Assume $m \geq 3$ and that the proposition is proven for m.
By Theorem 2.6 the automorphism group of Λ_{m+1} acts transitively on $S(\Lambda_{m+1})$ so we may choose a suitable $x_0 \in S(\Lambda_{m+1})$ and prove that

$$\sum_{x\in S(\Lambda_{m+1})}(x,x_0)^6 = c_{m+1}(\min(\Lambda_{m+1})).$$

To this aim we choose

$$x_0 = \tau(d_0) \in S(\Lambda_{m+1}) \text{ for some } d_0 \in S(\Delta_m).$$

Then by the description of $S(\Lambda_{m+1})$ in Lemma 3.5

$$(\star)\quad \sum_{x\in S(\Lambda_{m+1})}(x,x_0)^6 = 2^{m+1}\sum_{\ell\in S(\Lambda_m)}(\ell,d_0)^6 + \sum_{d\in S(\Delta_m)}(d,d_0)^6.$$

By induction hypotheses $S(\Lambda_m)$ and $S(\Delta_m)$ both form spherical 6-designs and hence the condition (a) from Remark 3.6 computes

$$\sum_{\ell\in S(\Lambda_m)}(\ell,d_0)^6 = c_m(\min(\Delta_m))$$

and

$$\sum_{d \in S(\Delta_m)} (d, d_0)^6 = 8c_m(\min(\Delta_m)).$$

Therefore $(\star)$ equals

$$(2^{m+1}+8)c_m(\min(\Delta_m)) = 2^3(2^{m-2}+1)s_m 2^{3m} 2^{3m-3} \frac{15}{2^m(2^m+2)(2^m+4)} =$$

$$s_m 2^{6m} \frac{15(2^{m-2}+1)}{2^{m+2}(2^m+2)(2^{m-2}+1)} = s_{m+1} 2^{6m} \frac{15}{2^{m+2}(2^m+2)(2^{m+1}+2)} =$$

$$s_{m+1} 2^{6m} \frac{15}{2^{m+1}(2^{m+1}+4)(2^{m+1}+2)} = c_{m+1}(\min(\Lambda_{m+1})).$$

As Λ_{m+1} and Δ_{m+1} are similar, also $S(\Delta_{m+1})$ is a 6-design. □

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